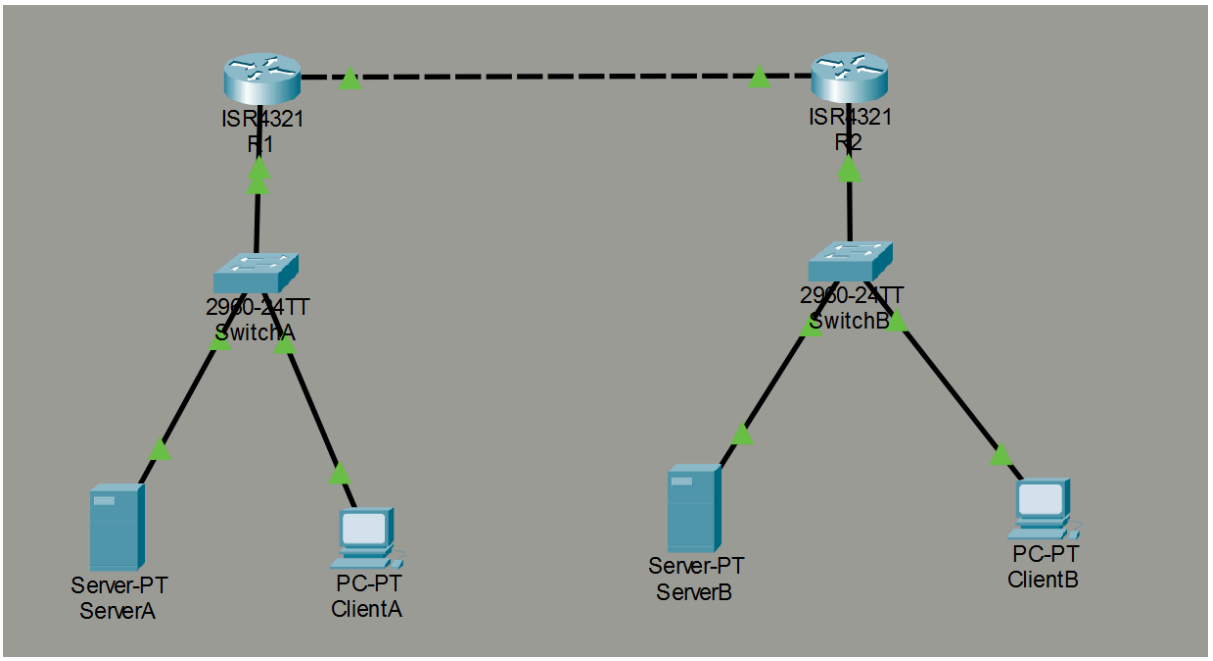

สมาชิกกลุ่ม

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Lab 4 using your specific repository, layout, and networking assumptions. This version **treats Local subnets as private networks** and **treats the remote server / inter-site network as a 10.10.x.x "internet/public network"** for the purposes of NAT and routing logic. It also assumes the repository hierarchy:

```
None
automation/
├── mockup-infra/
└── week02-stateless-stateful/
    └── phase1-mockup/
```

We will produce:

✅ A revised **high-fidelity lab plan** ✅ **Detailed addressing & topologies** ✅ **Cisco-themed network section** ✅ **IaC / VS Code section** ✅ **Troubleshooting and engineering reflection** ✅ A complete **README.md** ready to be committed to the repo

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🧪 LAB 4 Simulated Internet (10.10.0.0/16) & Private LANs with Stateful vs Stateless Services

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🎯 LAB GOALS

1. **Network Setup:** Create two local labs (A & B) connected via a simulated "public/internet" network (10.10.0.0/16)
2. **NAT + Routing:** Configure static routes and NAT so local traffic reaches remote services
3. **Application Deployment:** Use `mockup-infra` from your repository to deploy stateless and stateful APIs
4. **Testing:** Compare how sessions behave locally vs across the simulated internet

---

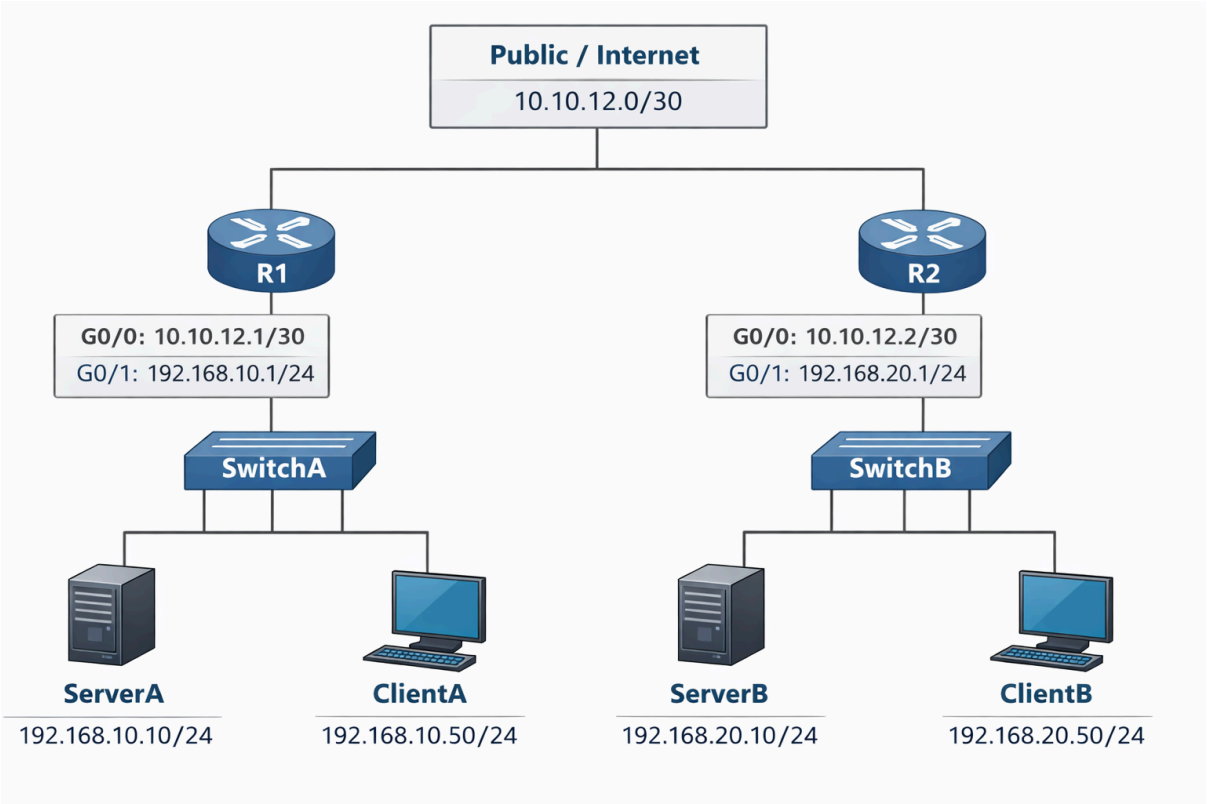
👥 TEAM BREAKDOWN (5–6 students)

Role	Count	Focus
Network Architects	2	Router configs, Cisco CLI
IaC/DevOps	2	VS Code, mockup-infra
QA/Test	1–2	Server/Client scripting, session behavior

🗺️ UPDATED NETWORK TOPOLOGY

```python
                10.10.12.0/30 "Public/Internet"
            ┌──────────────────────────┐
            │                          │
        R1-G0/0 (10.10.12.1)      R2-G0/0 (10.10.12.2)
            │                          │
   Private LAN A (192.168.10.0/24)     Private LAN B (192.168.20.0/24)
            │                          │
        SwitchA                      SwitchB
            │                          │
 ServerA (mockup)           ServerB (mockup)
 ClientA                        ClientB
```

We treat 10.10.12.0/30 as a **simulated internet** for the purposes of NAT + public reachability.



📍 ADDRESS TABLE

Component	Interface	IP	Mask	Role
R1	G0/0	10.10.12.1	/30	Public
R1	G0/1	192.168.10.1	/24	LAN A Gateway
ServerA	eth0	192.168.10.10	/24	Mockup infra A
ClientA	eth0	192.168.10.50	/24	Tester
R2	G0/0	10.10.12.2	/30	Public
R2	G0/1	192.168.20.1	/24	LAN B Gateway
ServerB	eth0	192.168.20.10	/24	Mockup infra B
ClientB	eth0	192.168.20.50	/24	Tester

🔧 PART 1 — CISCO-STYLE NETWORK CONFIGURATION

We will configure static routing + NAT to allow cross-site reachability.

🧱 R1 CONFIGURATION (LAN A)

```
None
enable
conf t

hostname R1

interface g0/0/0
 ip address 10.10.12.1 255.255.255.252
 ip nat outside
 no shutdown

interface g0/0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.20.0 255.255.255.0 10.10.12.2

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0/0 overload

end
write memory
```

```
R1(config)#interface g0/0/0
R1(config-if)#ip address 10.10.12.1 255.255.255.252
R1(config-if)#ip nat outside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

R1(config-if)#interface g0/0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

R1(config-if)#ip route 192.168.20.0 255.255.255.0 10.10.12.2
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
%Invalid interface number (Slot is empty)
R1(config)#ip nat inside source list 10 interface g0/0/0 overload
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
```

🧱 R2 CONFIGURATION (LAN B)

```
None
enable
conf t

hostname R2

interface g0/0/0
 ip address 10.10.12.2 255.255.255.252
 ip nat outside
 no shutdown

interface g0/0/1
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.10.0 255.255.255.0 10.10.12.1

access-list 10 permit 192.168.20.0 0.0.0.255
ip nat inside source list 10 interface g0/0/0 overload

end
write memory
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/0/0
R2(config-if)#ip address 10.10.12.2 255.255.255.252
R2(config-if)#ip nat outside
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R2(config-if)#interface g0/0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#ip nat inside
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

R2(config-if)#ip route 192.168.10.0 255.255.255.0 10.10.12.1
R2(config)#access-list 10 permit 192.168.20.0 0.0.0.255
R2(config)#ip nat inside source list 10 interface g0/0/0 overload
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
R2#
```

🧪 ROUTING & NAT VERIFICATION

On R1 & R2:

```
None
show ip route
show ip nat translations
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/0/0
L       10.10.12.1/32 is directly connected, GigabitEthernet0/0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
S    192.168.20.0/24 [1/0] via 10.10.12.2

R1#show ip nat translations
R1#
```

From ClientA:

```
None
ping 192.168.20.10  ← Remote server reachable
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::202:16FF:FE4C:71C7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.50
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

💻 PART 2 — IaC + MOCKUP-INFRA DEPLOYMENT (mockup-infra/README.md)

We will now set up mockup infra using the paths you specified:

```
None
automation/mockup-infra
automation/week02-stateless-stateful/phase1-mockup
```

🛠️ CLONE REPO

```shell
git clone https://github.com/boonsup/automation.git
cd automation
```

```
PS D:\งาน\network 5_3_2569> cd automation
PS D:\งาน\network 5_3_2569\automation>
```

🧰 VS CODE — OPEN PROJECT

In VS Code, open the `automation` folder.

📄 CREATE SERVER-SPECIFIC ENV FILES

ServerA (`ServerA.env`)

```
None
# Private LAN A
PUBLIC_NET_SUBNET=192.168.10.0/24
HOST_IP=192.168.10.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```

ServerB (`ServerB.env`)

```
None
# Private LAN B
PUBLIC_NET_SUBNET=192.168.20.0/24
HOST_IP=192.168.20.10
```

```
REMOTE_NET_PUBLIC=10.10.12.0/30
```

🚀 DEPLOY MOCKUP-INFRA

PowerShell (Windows Admin)

```None
cd mockup-infra
python manage.py init
python manage.py deploy --profile week02-stateless-stateful
```

Bash (Linux)

```Shell
cd mockup-infra
chmod +x scripts/*.sh
./scripts/init-env.sh
python manage.py deploy --profile week02-stateless-stateful
```

🟡 VERIFY CONTAINERS

```None
podman ps
```

Should show:

Container	Port
stateless-api	3000
stateful-api	3001

🧪 PART 3 — TESTING STATELESS vs STATEFUL (week03-stateless-stateful/phase1-mockup/README.md)

⚫ LOCAL TESTS (Private LAN)

From ClientA:

```shell
curl http://192.168.10.10:3000/info
```

→ Stateless

From ClientA:

```shell
curl -X POST http://192.168.10.10:3001/session
```

Store the returned `Session-ID`, then:

```shell
curl http://192.168.10.10:3001/data -H "Session-ID: <id>"
```

🔵 REMOTE TESTS (Via Simulated Internet)

From ClientA:

```none
curl http://10.10.12.2:3000/info
```

That will target **ServerB stateless** — NAT + routing should deliver

---

Session Test:

```
None
curl -X POST http://10.10.12.2:3001/session
```

→ ServerB returns a session ID

Then:

```
None
curl http://10.10.12.2:3001/data -H "Session-ID: <id>"
```

Check session persists across NAT

⚠️ TROUBLESHOOTING

❗ Cannot Reach Remote

- Gateway misconfiguration
- NAT ACL incorrect
- Wrong default gateway on clients

Check:

```
None
show ip route
ping 10.10.12.2 from R1
```

❗ Sessions Lose State

- Missing header
- Wrong container port
- Firewall blocking port 3001

🧠 ENGINEERING REFLECTION

1. **Stateless services don't depend on prior data**
2. **Stateful services require session headers**
3. NAT shouldn't affect logical session identity

📄 LAB README.md (for repository)

```
None
# Lab 4 — Stateful vs Stateless Across Public & Private Subnets

## Architecture
Two LANs connected via a simulated internet (10.10.12.0/30)
- LAN A: 192.168.10.0/24
- LAN B: 192.168.20.0/24
Routers provide static routing + NAT.

## Steps

### 1. Setup Network
Configure R1 and R2:
- Static routes
- NAT overload
- Verify ping

### 2. Deploy IaC
Configure `ServerA.env` and `ServerB.env`
Deploy via VS Code, PowerShell or Bash using:
```

python manage.py deploy --profile week02-stateless-stateful

```
None

### 3. Validate
- Local stateless vs stateful
- Remote stateless vs stateful
- Test session persistence
- Capture NAT translations

## Troubleshooting
- Check interface status, NAT ACLs
- Verify client gateways
```

```
## Extensions
- OSPF instead of static routing
- Load balancer in front of APIs
```